PumasAI DeepPumas

Joining models

Information transfer across endpoints — without ever fitting them together

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THE NEED

Use the biomarkers to sharpen the endpoint

A RECURRING ASK

Nearly half of our projects want **longitudinal biomarkers to improve a primary-endpoint prediction**. AST, ALT, imaging, dozens of markers.

The textbook answer: a **joint NLME model** with correlated random effects across all endpoints.

...BUT THE JOINT FIT IS A WALL

- 20 biomarkers → 40+ random effects, thousands of parameters. **FOCE will not fit that.** Even 4 endpoints gets hard.
- Tangled, slow, hard-to-iterate workflows.
- **Brittle** once neural components enter the structural model.

THE COMPROMISE

Fit each endpoint on its own — and pay one price

Fit the endpoints **independently** (or conditionally independently, given PK). Suddenly everything is easy: fast, plannable, parallelisable across a team, robust.

THE COST

Independent fits assume **independent priors** over the random effects — even though the true effects are **coupled**. You have thrown away the cross-endpoint dependence.

THE WHOLE TRICK

That discarded dependence is a **misspecified prior** — and the prior is the one thing we can fix **after the fact**, without touching the structural models.

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Four assumptions separate the joint likelihood — we relax one

Start from the joint marginal likelihood of two endpoints X, Y — the same integral as before, now over **both**:

$$p(Y, X \mid \theta) = \int p(Y, X \mid \eta, \theta)\, p(\eta \mid \theta)\; \mathrm{d}\eta$$

A1 · conditional independence given the random effects η

$$= \int p(Y \mid \eta, \theta)\, p(X \mid \eta, \theta)\, p(\eta \mid \theta)\; \mathrm{d}\eta$$

A2, A3 · separate parameters $\theta = (\theta_Y, \theta_X)$ and latent spaces $\eta = (\eta_Y, \eta_X)$

$$= \iint p(Y \mid \eta_Y, \theta_Y)\, p(X \mid \eta_X, \theta_X)\, p(\eta_Y, \eta_X \mid \theta)\; \mathrm{d}\eta_Y\, \mathrm{d}\eta_X$$

A4 · PRIOR INDEPENDENCE — THE ONLY ONE WE RELAX

Assume $p(\eta_Y, \eta_X \mid \theta) = p(\eta_Y \mid \theta_Y)\, p(\eta_X \mid \theta_X)$ and the integral **factorises completely** into two models you can fit in isolation:

$$p(Y, X \mid \theta) = p(Y \mid \theta_Y) \cdot p(X \mid \theta_X)$$

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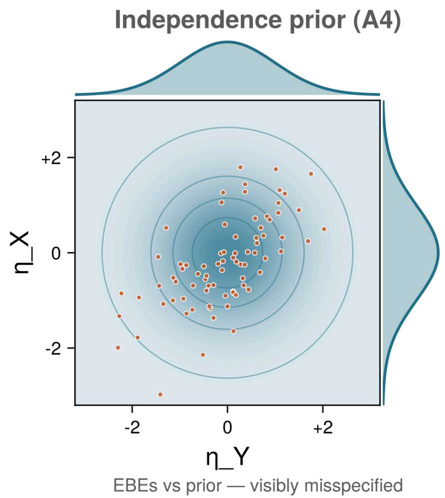
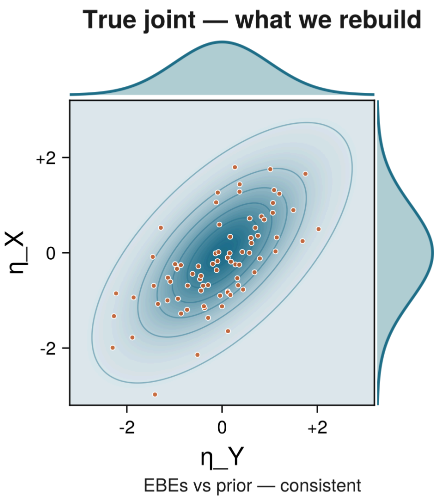
The data carry the coupling — even when the prior doesn't





LEFT — WHAT A4 ASSUMES

Marginals × marginals: circular, no coupling between η_X and η_Y.

RIGHT — WHAT THE DATA CARRY

Tilt and coupling. Each patient's **posteriors** already show it. That gap is the misspecification we rectify.

THE MOVE

Re-fit one joint prior on the per-patient posteriors

The coupling shows up **consistently** in the per-patient posteriors — so it belongs in the **prior**. Fit a flexible density p_φ to the joint those posteriors imply:

$$\underbrace{\tilde{p}(\eta_Y, \eta_X)}_{\text{target}} = \mathbb{E}_{(Y,X)}\left[\underbrace{p(\eta_Y \mid Y, \theta_Y)}_{\text{independent fit}}\,\underbrace{p(\eta_X \mid X, \theta_X)}_{\text{independent fit}}\right]$$

Each factor is independent by design — yet $\tilde{p}$ is **correlated**, because both observations come from the **same patient**.

Replace the independent priors with $p_\varphi \approx \tilde{p}$ — one call, the structural models left untouched:

- `replace_randeffs_dist(...)`

TWO DENSITY FAMILIES

Gaussian	linear cross-correlation only
Normalizing Flow	non-linear, non-Gaussian; no family to specify

Same supervision, different summary of it.

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THE RECIPE

Four steps, structural models untouched

1. **Fit each endpoint independently.** Share structure first (e.g. fit PK, carry its typical values into the PD models as fixed).
2. **Compute per-patient Laplace posteriors** of the random effects on the training subjects.
3. **Fit a joint prior** $p_{\varphi(\boldsymbol{\eta})}$ on those posteriors — a Gaussian, or a Normalizing Flow.
4. **Plug the learned prior back in** and evaluate on held-out test subjects.

NO JOINT FIT, EVER

The structural models, observation models, and θ never change. Only the prior over the random effects is replaced.

WHY JOIN POST-HOC

The workflow win is the real win

PLANNABLE

Each endpoint owns its model, data, diagnostics. Joining is a separate, restartable step.

SCALABLE — AND THE REASON WE BUILT IT

Joint fits with neural components are slow and initialisation-sensitive. Modular joining sidesteps that. (DeepNLME.)

REUSABLE

Re-couple a model from a past study without touching its structure — the flow can even fix a mildly misspecified prior.

MIXED FAMILIES

Couple an NLME with **any** latent-variable model that exposes a tractable posterior.

A **correlational** bridge — within-population dependence, not causal. And it is already in DeepPumas, today.

YOUR TURN

Let an early biomarker predict a late endpoint

One drug, two endpoints — a **fast biomarker** (sampled early, densely) and a **slow clinical endpoint** (sampled late, sparsely). In the exercise you:

1. fit an IDR model to **each endpoint independently**,
2. scatter the per-subject **EBEs** — the correlation is already in the data,
3. **join** them with `replace_randeffs_dist`,
4. predict a new patient's **clinical endpoint from the biomarker alone**.

THE TAKEAWAY

Nothing about the two models changed. The coupling was **in the data all along** — joining moves it into the prior, where predictions can use it.

WANT THE FULL BENCHMARKS?

How close post-hoc joining gets to the **full joint fit**, and where a **flow** beats a Gaussian on non-Gaussian structure — see the **poster**.